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ChipLayoutBench

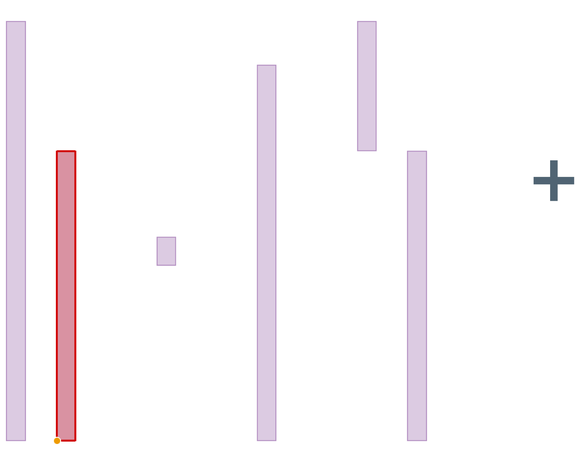
A Multimodal Benchmark for Vision-Language Models in Semiconductor Layout Understanding

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1 The number is in the text. Can the VLM find its polygon?

Question: What is the x-coordinate of the highlighted vertex?

BOTH INPUTS ARE PROVIDED TO THE MODEL



Layout image: highlighted target

GDS-derived text (excerpt)

poly_00001 | neighboring polygon
LAYER 13 DATATYPE 0
(320.78, 257.533) ...

→ **Wrong visual-to-text match**
Model selects poly_00001
returns x = 320.78

poly_00002 | actual target
LAYER 13 DATATYPE 0
(320.97, 255.485) ...

→ Target is poly_00002
correct x = 320.97

Failure: the model retrieves a real coordinate from the wrong polygon record.

Actual multimodal error, Appendix Fig. 29. Both numerical values are already present in the supplied GDS text.

2 Same layout, three ways to represent it

2,115 layouts · 6,808 test questions

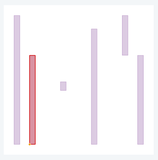
Real GDSII → sampled regions → template-based QA
Answers computed from geometry; shape labels checked by humans

TEXT
GDS geometry serialized as text

```
POLYGON poly_00002
LAYER 13  DATATYPE 0
(320.97, 255.485) ...
```

Polygon IDs + layer / datatype
+ exact vertex coordinates

IMAGE + METADATA
Rasterized semiconductor layout



Spatial metadata:
Reference coordinate
Scale: 1 nm / pixel
Layer-to-color mapping

Pixels + spatial reference, without full vertex lists

BOTH
The text AND image of the same layout

GDS text
IDs · layers
coordinates

+



+ spatial metadata

Match visual targets to exact polygon records

What do we ask the models?

5 task categories · Example Q&A from Table 1

Count & describe
Recognition & description
How many polygons have 4 vertices?
→ **4**

Find & retrieve
Localization & retrieval
Which polygon is closest to the top-left corner?
→ **poly_00042**

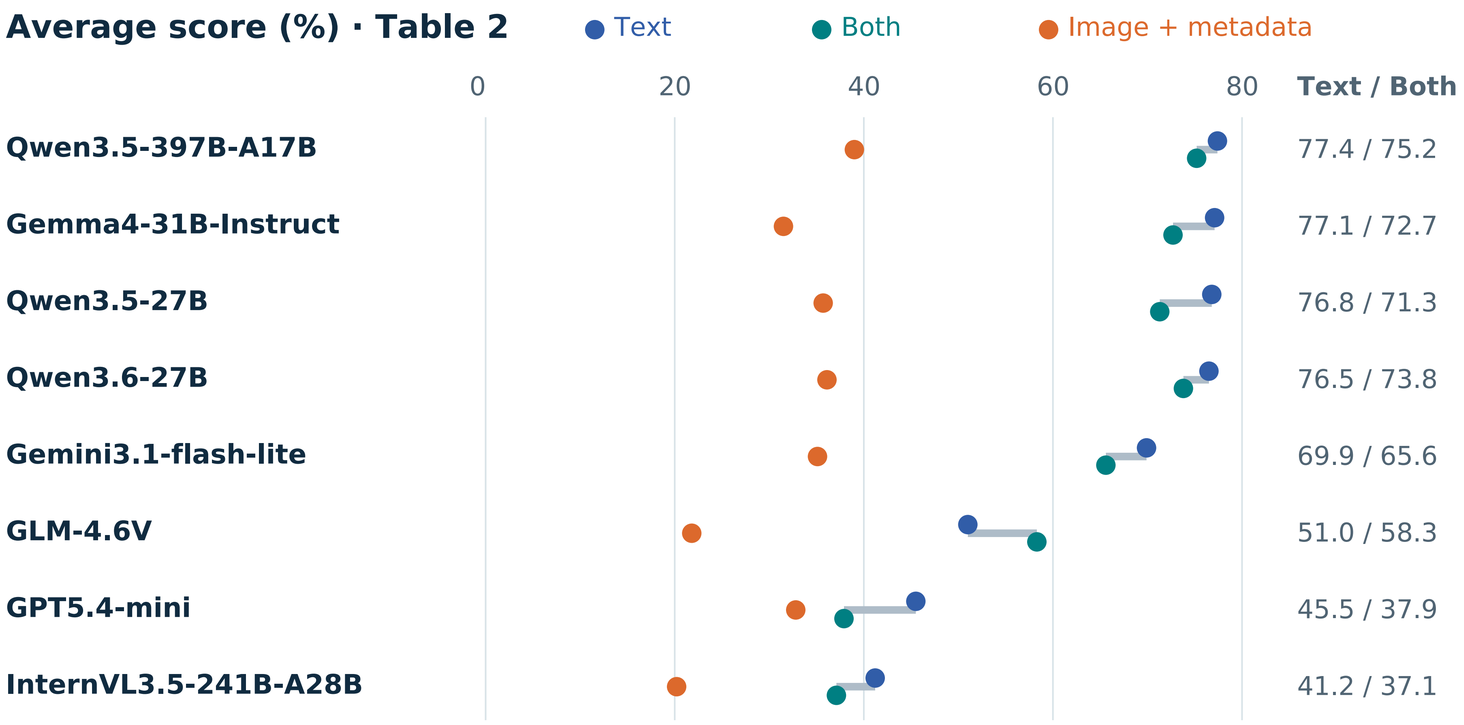
Measure & reason
Geometric & spatial reasoning
What is the extruded area of poly_00038?
→ **1.5314**

Check layout quality
DRC & hotspot detection
How many m1 width-rule violations (0.018 µm)?
→ **3**

Match image to text
VLM-specific evaluation
Which polygon ID matches the highlighted polygon?
→ **poly_00082**

3 Adding an image often reduces performance

Average score (%) · Table 2



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models score lower with both modalities than with text alone.

Even when images may help

56.2% → 52.3%
Text Both

Macro-average exact match on image-helpful questions (Table 3). These are still answerable from text.

What the benchmark reveals

Adding visual context does not reliably improve exact layout reasoning.

The failure case illustrates a key challenge: linking the visual target to the correct GDS record.

Oct 25 · 11:00–12:30 · Hall H · Poster A